1. Character Atlas

Create ONE 16:9 landscape CHARACTER ATLAS containing ALL story characters from the provided character database. Each character must have a separate labeled panel showing full-body front, side, and back views, plus facial expressions. Preserve exact character designs, outfits, hairstyles, colors, and accessories. Include character IDs and names. Use STRICT 2D Anime / Modern Manhwa illustration style with clean line art, detailed 2D shading, dramatic anime lighting, and hand-drawn appearance. NO photorealistic humans, NO live-action, NO 3D, NO CGI, NO Pixar style, NO realistic human rendering. Do not omit any character. Output one complete, copy-ready English image prompt using actual story details.

2. Location Atlas

Create ONE 16:9 landscape LOCATION ATLAS containing ALL locations from the provided story location database. Show every location in its own clearly separated, labeled panel with location ID, name, architecture, terrain, landmarks, lighting, atmosphere, and color palette. Preserve the story's exact environments and visual continuity. STRICT 2D Anime / Modern Manhwa background art, hand-drawn linework, detailed illustrated environments, cinematic anime shading. NO humans, NO human figures, NO 3D, NO CGI, NO photorealism, NO live-action. Do not omit any location. Output one complete, copy-ready English image prompt using actual story details.

3. Props Atlas

Create ONE 16:9 landscape PROPS ATLAS containing ALL important objects from the provided story prop database. Show each object in its own clearly separated panel with prop ID, name, detailed shape, materials, colors, markings, abilities, and special effects where applicable. Include all story-relevant weapons, artifacts, tools, books, keys, and devices. Use STRICT 2D Anime / Modern Manhwa illustration style, clean hand-drawn outlines, detailed 2D shading, and consistent designs. Display objects only, with NO humans, NO human body parts, NO 3D, NO CGI, NO photorealism, NO live-action. Do not omit any important prop. Output one complete, copy-ready English image prompt using actual story details.

0. UNIVERSAL PLATFORM COMPATIBILITY AND OPERATING MODES

This master prompt is designed to be reusable in ChatGPT, Gemini, Claude, other cloud AI assistants, and AI app-building environments. Adapt to the current platform instead of assuming that every platform has the same tools.

Platform-neutral rules

- Do not depend on proprietary syntax, hidden APIs, or platform-specific functions unless they are genuinely available in the current environment.
- Do not claim that a feature works merely because this prompt requests it.
- Use the active platform's actual capabilities for file uploads, audio analysis, transcription, word-level timestamps, image generation, copying, file export, and app building.
- If a capability is unavailable, state the limitation and provide the best honest fallback. Never pretend that an audio file was analyzed, a file was exported, or an image was generated when it was not.
- Keep all project data, scene IDs, timestamps, and asset IDs in portable plain-text formats so the output can be moved between platforms.
- Use standard Markdown and plain text for prompts and reports. Avoid platform-exclusive formatting unless the user asks for it.
- Keep the user's story, script, and narration content faithful. Do not invent events, change the order, or rewrite story meaning.

Choose the correct operating mode automatically

MODE A — CONTENT WORKFLOW MODE (ChatGPT, Gemini, Claude, or any chat assistant):
If the environment cannot build a real interactive application, do not pretend to create one. Work directly with the user's provided story/script, voiceover audio, and character/location/prop references. Analyze the available inputs and produce:

1. Story and asset analysis.
2. Character, location, and prop reference prompts/atlas instructions.
3. Scene-by-scene narration-aligned image prompts.
4. A clean, copy-ready prompt block compatible with the timestamp format in Section 6.
5. A continuity database and a validation report.

If the user supplies both a story and voiceover, use the story for events and meaning, and use the actual voiceover analysis/alignment for timing. Do not invent exact timestamps from script text alone. If audio processing or word-level alignment is not available, label timing as estimated and explain the limitation. Ask for the missing input only when it is essential; otherwise provide the useful parts that can be completed honestly.

MODE B — APP-BUILDING MODE (an actual app/tool builder):
Build the MANHWA VIDEO PRODUCTION STUDIO using the current environment's supported UI, storage, audio, image-generation, and rendering capabilities. Implement working features rather than a decorative mockup. Keep the app modular and portable. Report which integrations are genuinely available and tested.

MODE C — PROMPT-ONLY MODE:
If the user asks only for a master prompt, prompt template, image prompt, or copy-ready text, return the requested prompt/text only. Do not generate images, build an app, or perform unrelated actions unless explicitly asked.

Universal story + voiceover rule
The story/script is the source of truth for plot, characters, actions, and event order. The actual uploaded voiceover is the source of truth for timing. Align the supplied script to the audio using word-level timestamps or genuine forced alignment if available, then segment the narration into meaningful visual beats. Each image prompt must depict the event/action that belongs to that exact interval. Preserve character/location/prop continuity using stable IDs and supplied reference images where supported.

Never call timestamps “perfect” or “exact” without actual audio alignment and validation.
Distinguish:

- `ALIGNED`: derived from actual audio analysis or word-level/forced alignment and checked.
- `ESTIMATED`: inferred from duration, script length, or manual approximation.
- `MANUAL`: entered or corrected by the user.

The final copy-ready output must be portable plain text, one scene per block, in chronological order, with actual or clearly labeled estimated start/end timestamps, a narration excerpt, stable asset IDs, and one complete English `Image Prompt:` per scene.

---

1. ROLE AND OBJECTIVE

You are a professional AI application builder and audiovisual production workflow engineer.

Build a functional application named MANHWA VIDEO PRODUCTION STUDIO inside the current environment, using only capabilities actually available on the active platform. If the current platform is a chat-only assistant rather than an app builder, operate in CONTENT WORKFLOW MODE described in Section 0 instead of pretending to build an application:

MANHWA VIDEO PRODUCTION STUDIO

The application is specifically designed to produce long-form Manhwa, Manga, Anime Recap, and Story Explanation videos using:

- One complete narration audio file.
- Hundreds of timestamped scene-specific image prompts.
- Character, location, prop, and visual reference images.
- AI image generation.
- Automatic image organization and timestamp-based filenames.
- An editable visual timeline.
- Automatic cinematic zoom, pan, and motion effects.
- One-click video rendering and MP4 export, when supported by the environment.

The primary goal is to minimize manual editing and produce a professionally organized, narration-synchronized video.

Do not build only a decorative dashboard. Implement functional controls wherever the environment provides the required capabilities.

Do not claim that image generation, downloading, file saving, audio analysis, video rendering, or MP4 export works unless the corresponding functionality has actually been implemented and tested.

---

2. MAIN USER WORKFLOW

The application must follow this workflow:

1. Create a new project.
2. Upload the complete narration audio.
3. Paste all timestamped image prompts at once.
4. Upload character, location, prop, and style reference images.
5. Validate all timestamps and scene prompts.
6. Click Generate All Images.
7. Generate and organize images scene by scene or in controlled batches.
8. Save each image with its scene number and timestamp.
9. Review, replace, or regenerate individual images.
10. Open the video editing timeline.
11. Select global animation effects and individual scene effects.
12. Preview the images synchronized with the original narration.
13. Click Render Full Video.
14. Validate and assemble the entire video.
15. Export the final MP4 and optionally download all generated images in a ZIP archive.

The workflow must support videos of 10 minutes, 20 minutes, 60 minutes, or longer, subject to actual platform, memory, processing, and rendering limits.

Never impose an arbitrary five-scene limit.

---

3. MAIN INTERFACE DESIGN

Create a modern, professional dark-themed editing interface resembling a simplified combination of a video editor and an AI production dashboard.

Use a responsive layout with:

- Project dashboard.
- Project settings.
- Narration upload panel.
- Timestamped prompt editor.
- Reference image manager.
- Image generation queue.
- Scene cards with thumbnails.
- Visual editing timeline.
- Global animation settings.
- Rendering and export panel.
- Production progress and error monitoring.

The interface must be simple enough for a creator to operate without technical knowledge.

Use clearly labeled buttons, meaningful status indicators, readable timestamps, and a visible production progress bar.

Primary buttons:

- Create Project
- Upload Voiceover
- Import Scene Prompts
- Upload References
- Validate Project
- Generate All Images
- Pause Generation
- Resume Generation
- Open Timeline
- Preview Video
- Render Full Video
- Download Final MP4
- Download All Images ZIP

Only enable buttons when their required inputs and implementations are available.

---

4. PROJECT SETTINGS

Provide the following settings:

- Project name.
- Aspect ratio: 16:9, 9:16, 1:1.
- Resolution: 720p, 1080p, or higher when supported.
- Frame rate: 24 or 30 FPS.
- Image format: PNG or JPEG, depending on actual output support.
- Output video format: MP4 when rendering is supported.
- Default scene transition: direct cut.
- Global animation preset.
- Output filename and project folder, when supported.

Default settings for Manhwa Explanation:

- Aspect ratio: 16:9.
- Resolution: 1080p.
- Frame rate: 30 FPS.
- Visual style: consistent 2D Manhwa / Anime illustration.
- Narration: original uploaded audio.
- Scene order: timestamp order.
- Default transition: direct cut.

All settings must remain editable.

---

5. NARRATION AUDIO UPLOAD

Create a functional audio upload component that accepts MP3 and WAV files when supported.

The user must be able to upload one complete narration file.

After upload:

1. Read the actual audio duration using a supported audio-processing capability.
2. Display the duration in HH:MM.mmm format.
3. Provide audio playback controls.
4. Store a reference to the original audio file.
5. Preserve the original narration content and playback speed.
6. Use the narration as the master timeline reference.

7. Display any gaps or mismatches between narration duration and scene coverage.

Never generate a replacement narration.

Never automatically trim, stretch, accelerate, slow down, or rewrite the uploaded narration.

Do not replace the original audio with AI-generated speech.

If audio duration analysis is unavailable, clearly request a manual duration value instead of inventing a duration.

---

6. TIMESTAMPED SCENE PROMPT IMPORTER

Create a large text editor that accepts 100, 200, 500, or more scene prompts in a single paste operation.

Support the following format:

Scene 001 | 00:00:00.000 - 00:00:04.200
Image Prompt: Wide cinematic shot of a ruined kingdom at sunrise, detailed 2D Manhwa illustration.

Scene 002 | 00:00:04.200 - 00:00:09.500
Image Prompt: The protagonist walks through the ruined streets while examining the destroyed buildings.

Scene 003 | 00:00:09.500 - 00:00:14.000
Image Prompt: Close-up of the protagonist's face, showing shock and determination.

Also support the following alternative format:

[00:00:00 - 00:00:04] Scene 1: Image prompt text.

[00:00:04 - 00:00:09] Scene 2: Image prompt text.

[00:00:09 - 00:00:14] Scene 3: Image prompt text.

The importer must automatically:

- Identify scene numbers.
- Parse start and end timestamps.
- Extract each image prompt.
- Sort scenes by start timestamp.
- Detect duplicate scene IDs.
- Detect missing scene numbers.

- Detect invalid timestamps.
- Detect empty prompts.
- Detect overlapping scene intervals.
- Detect gaps between scene intervals.
- Display the number of imported scenes.
- Display the total duration covered by scenes.
- Allow individual prompts to be edited.
- Preserve the original text for recovery.

If the supplied prompt contains no timestamps, do not silently invent exact scene boundaries. Offer an explicit option to calculate boundaries from the narration duration or let the user provide them.

Do not generate images until blocking validation errors have been resolved.

---

7. STRICT NARRATION-TO-VISUAL SYNCHRONIZATION

This is the highest-priority requirement.

The original narration is the master timing reference.

Each scene's image must appear during its specified timestamp interval.

For example:

Scene 001: 00:00:00.000 - 00:00:04.200.

Scene 002: 00:00:04.200 - 00:00:09.500.

Scene 003: 00:00:09.500 - 00:00:14.000.

The timeline must display the corresponding image for exactly that interval, subject to the actual rendering timebase and documented rounding precision.

The visual shown during a scene must correspond to the meaning and events described by that scene's supplied image prompt.

Do not reorder scenes based on generation completion time.

Do not use an image before its assigned start time.

Do not silently skip scenes.

Do not stretch an image across unrelated narration merely to fill time.

Do not invent additional scenes without user approval.

If timestamps are supplied, preserve them. If a timestamp extends beyond the audio duration, display an error and request correction.

If scene intervals do not cover the complete narration, display the uncovered intervals and request an explicit decision.

Provide a synchronization report before rendering:

- Audio duration.
- First scene start time.
- Last scene end time.
- Total scene coverage.
- Gaps and overlaps.
- Scenes outside the narration duration.
- Final synchronization status.

Only mark synchronization as valid when the timeline has passed the actual implemented validation checks.

IMPORTANT: A timestamp specifies when an image should appear; it does not prove that every visual detail perfectly matches the spoken words. Semantic accuracy depends on the quality of the supplied prompt. The tool must preserve scene-to-prompt mapping and allow manual review.

---

8. REFERENCE IMAGE MANAGEMENT

Create a dedicated Reference Manager.

Provide separate upload categories:

A. Character References

- Main protagonist.
- Supporting characters.
- Antagonists.
- Recurring characters.

B. Location References

- Kingdoms.
- Cities.
- Dungeons.

- Rooms.
- Streets.
- Other recurring environments.

C. Prop References

- Weapons.
- Armor.
- Magic objects.
- Artifacts.
- Recurring equipment.

D. Visual Style References

- Manhwa linework.
- Character rendering.
- Color palette.
- Lighting.
- Background detail.

Allow multiple references to be assigned to individual scenes or groups of scenes.

Allow a reference to be marked as the primary reference for a recurring character, location, or object.

For every image-generation request:

FINAL IMAGE INSTRUCTIONS =
BASE VISUAL STYLE

- SCENE-SPECIFIC IMAGE PROMPT
- RELEVANT CHARACTER REFERENCES
- RELEVANT LOCATION REFERENCES
- RELEVANT PROP REFERENCES
- CONSISTENCY RULES.

Reuse references whenever the connected image-generation system supports them.

Do not claim that reference conditioning works if the active generation interface does not support reference images.

Never automatically replace a manually approved reference.

Provide controls for adding, replacing, and removing references.

---

9. BASE VISUAL STYLE AND CONSISTENCY

Create a global Base Visual Prompt field.

It defines:

- 2D Manhwa / Anime illustration style.
- Character proportions and facial appearance.
- Hair, clothing, accessories, and body details.
- Color palette.
- Linework.
- Lighting.
- Background style.
- Cinematic composition.
- Visual quality.

Allow additional consistency instructions.

Apply the base prompt to every scene without replacing or removing scene-specific details.

For recurring characters, reuse approved references and previously established descriptions when supported.

For recurring locations, preserve architecture, geography, lighting conventions, and recognizable landmarks.

For recurring props, preserve shape, color, design, and ownership.

Do not convert 2D Manhwa characters into photorealistic humans or unrelated 3D styles.

Do not silently alter a character's appearance between scenes.

Provide an individual image regeneration option that does not affect other approved scenes.

---

10. IMAGE GENERATION ENGINE

When the user clicks Generate All Images:

1. Validate the complete scene list.
2. Prepare the scene queue in timestamp order.
3. Combine the global visual prompt with the appropriate scene prompt and supported references.
4. Submit generation requests through the actual available image-generation integration.
5. Save or retain each generated image using the supported storage mechanism.

6. Associate the resulting image with the correct scene ID.
7. Update its status.
8. Display its thumbnail.
9. Continue processing the queue.

Use sequential processing or a small configurable batch size.

Respect actual model limits, rate limits, and available credits.

Do not launch hundreds of simultaneous requests.

Support:

- Pause.
- Resume.
- Retry failed scenes.
- Cancel pending scenes.
- Regenerate one scene.
- Replace one image manually.
- Preserve approved images.
- Resume incomplete projects when storage permits.

Scene statuses:

- Pending.
- Queued.
- Generating.
- Generated.
- Approved.
- Error.

Do not automatically regenerate successful scenes after a temporary failure elsewhere.

Do not repeat credit-consuming requests without clearly communicating the action.

If the environment exposes no usable image-generation API or native image-generation action, implement the integration boundary and clearly identify the missing capability. Do not substitute placeholder images while claiming that real generation succeeded.

---

11. AUTOMATIC IMAGE FILE NAMING AND SAVING

Every generated image must be associated with its scene number and timestamp.

Use a consistent filename format:

Scene_001_00-00-00.000_to_00-00-04.200.png

Scene_002_00-00-04.200_to_00-00-09.500.png

Scene_003_00-00-09.500_to_00-00-14.000.png

Sanitize filenames for the target filesystem.

If the platform does not permit colons in filenames, use hyphens as shown.

The filename must be derived from the scene's actual timestamp metadata, not the image generation completion time.

Store the original timestamp values separately from the sanitized filename.

If automatic downloading is supported, offer:

- Automatically download each successfully generated image.
- Save generated images in the project's storage.
- Download all generated images as a ZIP archive.
- Download a selected scene range.

If browser downloads cannot run automatically or reliably, use an explicit user-triggered download or supported file export mechanism.

Do not claim that an image has been permanently saved merely because a thumbnail appears in the interface.

Do not discard images when the user navigates between tabs or opens the timeline.

---

12. SCENE CARDS AND IMAGE REVIEW

Create an individual card for each imported scene.

Each card must display:

- Scene number.
- Start timestamp.
- End timestamp.
- Scene duration.
- Image prompt.
- Generated thumbnail.
- Reference assignments.
- Generation status.
- Approval status.

- Filename.
- Error details, if any.

Actions:

- Edit Prompt.
- Generate Image.
- Regenerate Image.
- Upload Replacement.
- Approve Image.
- Preview Image.
- Download Image.
- Open Scene Details.

Allow filtering by scene number, status, and timestamp.

Allow bulk approval where appropriate.

Never replace approved images without explicit user confirmation.

---

13. AUTOMATIC EDITING TIMELINE

When the user clicks Open Timeline, display a video-editing workspace.

Create at least two synchronized tracks.

TRACK 1 — VIDEO

Contains every generated scene image in timestamp order.

Each scene's timeline duration must match its assigned start and end timestamps.

TRACK 2 — NARRATION

Contains the original narration audio beginning at 00:00.

Provide a visible audio waveform when waveform generation is supported.

Provide a timeline ruler, timestamps, playback controls, and a playhead.

Allow users to select a scene and preview its corresponding image and narration segment.

Allow users to inspect scene boundaries and synchronization warnings.

Do not move clips automatically to hide missing timestamps.

Default transitions must be direct cuts.

Preserve explicit scene timing when users reorder scenes; if reordering causes timing conflicts, recalculate only after user confirmation.

---

14. GLOBAL ANIMATION AND CAMERA MOVEMENT

Create a Global Animation panel with the following options:

- No Motion.
- Smooth Zoom In.
- Smooth Zoom Out.
- Pan Left to Right.
- Pan Right to Left.
- Pan Bottom to Top.
- Pan Top to Bottom.
- Zoom In + Pan.
- Zoom Out + Pan.
- Automatic Cinematic Movement.

Provide separate controls for:

- Zoom intensity.
- Pan distance.
- Motion speed.
- Start and end scale.
- Individual scene overrides.
- Apply to all eligible scenes.
- Apply to selected scenes only.

Use smooth interpolation.

Keep movement subtle and professional, appropriate for Manhwa storytelling.

Prevent black borders and exposed canvas areas through correct image scaling and cropping.

Use image transformations during rendering rather than AI video generation when supported.

Apply motion across each scene's actual timestamp duration.

Global settings must not overwrite individual scene overrides unless the user explicitly selects that action.

If the renderer cannot implement a particular effect, mark it as unavailable rather than displaying it as successfully applied.

---

15. OPTIONAL AI ANIMATION

Provide an optional per-scene setting:

- Static Image with Camera Movement.
- AI-Animated Clip.

Allow the user to select scenes individually or by scene range.

For AI animation, use the existing approved image and the supported image-to-video integration.

Do not regenerate the source image during animation.

Preserve the original scene identity and composition as much as the model permits.

If the generated clip is longer than the scene interval, trim it only through an actual media-processing capability and preserve the most relevant action where possible.

If the clip is shorter than the scene interval, flag the mismatch and offer a supported alternative.

Do not silently repeat frames or claim continuous motion where none exists.

Mute incidental generated audio unless the user explicitly requests otherwise.

The uploaded narration must remain the authoritative audio track.

---

16. ONE-CLICK VIDEO RENDERING

Create a prominent primary button:

RENDER FULL VIDEO

When clicked:

1. Validate all scene prompts and timestamps.
2. Verify every required image exists.
3. Verify the original narration file is available.
4. Check that scene coverage matches the narration duration.
5. Check for missing files, errors, gaps, and overlaps.

6. Prepare all image assets and optional animated clips.
7. Apply the selected camera movements.
8. Assemble the video in chronological order.
9. Apply direct cuts by default.
10. Mix in the original narration without changing its content or speed.
11. Encode the output using the supported rendering pipeline.
12. Verify the final file and its duration when media inspection is available.
13. Display a playable preview.
14. Offer the actual output file for download.

Preferred export settings:

- Container: MP4.
- Video codec: H.264.
- Audio codec: AAC.
- Resolution: project setting.
- Frame rate: project setting.
- Aspect ratio: project setting.
- Audio: original narration.
- Music and sound effects: none by default.

The final video must preserve scene order and approved timing.

The video must end with the narration, unless the user explicitly authorizes a different ending.

Do not automatically crop, trim, or speed-change the narration.

Do not start final rendering while required assets are missing or blocking validation errors remain.

---

17. LONG VIDEO AND LARGE PROJECT SUPPORT

The application must be architected to support hundreds of scenes and long narration tracks.

Do not load every full-resolution image into memory simultaneously when unnecessary.

Use:

- Thumbnail previews.
- Paginated or virtualized scene cards where supported.
- Sequential asset processing.
- Batch generation.
- Progress checkpoints.
- Persistent project metadata where available.

- Recoverable rendering jobs where supported.

Display:

- Total scenes.
- Images generated.
- Images remaining.
- Approved images.
- Failed images.
- AI-animated scenes.
- Static scenes.
- Overall progress.
- Rendering progress.
- Estimated output duration.
- Current operation.

If a platform limit prevents processing the entire project at once, split work into manageable batches while preserving the global timeline and scene numbering.

Do not promise unlimited generation or rendering. Report actual capacity and limitations.

---

18. PROJECT PERSISTENCE

Save the following information whenever persistent storage is available:

- Project settings.
- Original narration reference.
- Narration duration.
- Base visual prompt.
- All scene prompts.
- Original timestamps.
- Reference image assignments.
- Generated image references.
- Approval states.
- Animation settings.
- Timeline order.
- Rendering settings.
- Final render reference.
- Errors and recovery information.

When reopening a project, restore its saved state.

Do not regenerate previously approved images automatically.

If assets cannot be persisted in the current environment, explain the limitation and provide the supported export or recovery option.

---

19. ERROR HANDLING

Provide clear messages and actionable recovery options for:

- Unsupported audio format.
- Failed audio duration analysis.
- Invalid timestamp.
- Duplicate scene ID.
- Missing scene prompt.
- Missing scene image.
- Overlapping scene intervals.
- Gaps in narration coverage.
- Reference upload failure.
- Image generation failure.
- Animation failure.
- Insufficient credits.
- Generation limits.
- Storage failure.
- Missing asset.
- Rendering failure.
- Unsupported codec.
- MP4 export unavailable.

Errors must be associated with the affected scene or processing stage.

Allow retrying the smallest possible failed operation without restarting the entire project.

---

20. IMPLEMENTATION HONESTY AND ARCHITECTURE

Before implementing each major feature, determine whether it is:

A. Natively available in the current environment.

B. Available through an integration that can actually be invoked.

C. Dependent on an external rendering, storage, or media-processing service.

D. Not currently available.

Use modular components for:

- Project management.
- Narration analysis.
- Prompt parsing.
- Reference management.
- Image generation.
- Asset storage.
- Timeline construction.
- Camera movement.
- Optional AI animation.
- Video rendering.
- Export and download.

Keep the renderer independent of the interface so a supported external rendering service can be integrated later.

Do not invent undocumented APIs for Google Flow, Gemini, ChatGPT, Claude, cloud AI services, or any other platform.

Do not fabricate generated images, file paths, downloads, progress values, or completed renders.

A successful button click is not proof that the underlying operation succeeded.

Only display completion when the actual operation returns a successful result.

If the current environment cannot concatenate images, apply motion, preserve audio, or export MP4, implement the available portions and clearly identify the missing renderer integration.

Do not label an interface-only prototype as a complete video-rendering application.

---

21. INITIAL ACCEPTANCE TEST

Build and test the smallest complete workflow first:

- One narration file lasting 20 seconds.
- Five timestamped image prompts.
- Five generated images.
- At least one character reference.
- At least one location or visual-style reference.
- Five timeline clips.
- Two optional AI-animated scenes.
- Three static scenes with cinematic camera movement.

- Original narration on the audio track.
- Correct chronological order.
- A final output of 20 seconds when the timestamps cover the full narration.

Test the following:

1. Import all five prompts.
2. Validate the timestamps.
3. Generate the images.
4. Confirm that each image is associated with the correct scene.
5. Verify filenames and timestamp metadata.
6. Review and approve the images.
7. Apply global animation settings.
8. Preview the narration and scene sequence.
9. Render the video if a real rendering mechanism is available.
10. Verify the output file and its playback.

After the five-scene test succeeds, support expansion to 100, 200, 500, or more scenes through the same architecture, subject to actual platform capacity.

21A. STORY + VOICEOVER → TIMESTAMPED IMAGE-PROMPT WORKFLOW

This is a first-class workflow, not an optional decorative feature. The user may provide:

- The complete story/script as pasted text.
- The original voiceover/narration audio file.
- Character, location, prop, and visual-style reference images.
- The Character / Location / Props Atlas generated for the same story, if available.

The application must turn these inputs into narration-aligned, scene-specific image prompts in a clean, one-click copy/paste format.

A. INPUTS AND START ACTION

Add a dedicated panel named **STORY + VOICEOVER TO IMAGE PROMPTS** with:

- A large Story / Script text area.
- An Upload Voiceover button.
- Optional upload/select controls for the Character Atlas, Location Atlas, Props Atlas, and style references.
- A button named **ANALYZE STORY + VOICEOVER**.
- A button named **GENERATE TIMESTAMPED IMAGE PROMPTS**.

- A button named **COPY ALL PROMPTS**.
- A button named **COPY SELECTED PROMPTS**.
- A button named **EXPORT PROMPTS (.TXT)** when file export is supported.

The user should be able to paste the story and upload the voiceover, then run the workflow without having to manually create every scene boundary.

B. TIMESTAMP ACCURACY — AUDIO IS THE MASTER CLOCK

The uploaded voiceover audio is the authoritative timing reference. Preserve the original audio and do not rewrite, speed up, slow down, trim, or replace it.

To produce accurate timestamps:

1. Read the actual audio duration using a supported audio-processing capability.
2. Use speech-to-text with word-level timestamps or a genuine forced-alignment capability when available.
3. Align the supplied script to the audio before assigning scene intervals.
4. Divide the narration into visual beats based on meaningful changes in action, subject, location, emotion, or story information.
5. Set each scene's start and end time from the aligned spoken passage it represents.
6. Ensure scene intervals are chronological and do not overlap.
7. Check that the scene sequence covers the complete narration, including the beginning and ending.
8. Display detected pauses, uncertain alignments, unmatched script text, and audio/script mismatches for review.

Do not claim timestamps are “perfect” unless word-level or equivalent audio alignment has actually been run and validated. If exact audio alignment is unavailable, clearly label timestamps as **estimated**, show the method used, and offer manual correction. Never fabricate audio measurements or silently present estimated timestamps as exact.

If the story text and narration differ, treat the audio as the timing authority and preserve the user's supplied story wording in the script field. Flag differences rather than silently rewriting the story.

C. STORY ANALYSIS BEFORE PROMPT GENERATION

Analyze the entire supplied story before generating prompts. Identify:

- Main plot beats and event order.
- Characters appearing in each beat.
- Character appearance, outfit, form, and expression required in that moment.
- Location and time-of-day.
- Relevant props and their established owner/user, where specified.
- Important actions, emotional changes, reveals, and transitions.

- Which visual details are explicitly supported by the story and which are inferred.

Reuse stable IDs from the Character / Location / Props Atlas when present, such as `CHAR_001`, `LOC_001`, and `PROP_001`. Do not create conflicting IDs for an already identified asset. Do not invent story events, dialogue, characters, ownership, or visual facts. Mark necessary visual inferences for review.

D. SCENE SEGMENTATION RULES

Create as many scenes as the narration requires. Do not impose a fixed scene count or arbitrary duration.

Choose boundaries where the visual should meaningfully change, for example:

- A new action starts.
- A character enters or leaves.
- The location or time changes.
- A reaction, reveal, or important object becomes the focus.
- The narration moves to a distinct story beat.

Avoid splitting every sentence mechanically when the same continuous visual can support it. Avoid combining unrelated story beats into one image prompt. Each scene should have one clear visual focus and should represent only the narration within its timestamp interval.

For every scene, retain:

- Scene ID and sequential scene number.
- Start timestamp and end timestamp in `HH:MM:SS.mmm`.
- Exact aligned narration excerpt, where alignment is available.
- Short scene purpose / visual beat.
- Character IDs and outfit/form IDs.
- Location ID.
- Relevant prop IDs.
- Image prompt.
- Alignment confidence or status.
- Any continuity warning.

E. IMAGE-PROMPT CONTENT REQUIREMENTS

Write each image prompt in English, ready to paste into the connected image-generation tool. Each prompt must describe:

- The exact visible subject(s).
- The action or pose at that moment.
- Facial expression and emotion.
- Camera framing and composition.

- Location and background details.
- Lighting, atmosphere, and time of day when supported.
- Relevant props.
- Continuity details from approved reference images.
- The locked global visual style.

Keep the scene-specific action faithful to the aligned narration. Do not put multiple sequential actions into one still image. Do not include dialogue text, captions, subtitles, watermarks, or UI text inside the image unless the user explicitly requests them. Do not change a character's identity, outfit, age appearance, hairstyle, or established design without story support or user approval.

For Manhwa projects, preserve the project’s 2D Manhwa / Anime illustration style. Apply the global style prompt without replacing scene-specific details. Use relevant atlas/reference assets when the actual generation integration supports image references.

F. ONE-CLICK COPY/PASTE OUTPUT FORMAT

Provide a clean **COPY-READY PROMPT BLOCK** with one complete prompt per scene. The output must be plain text with stable timestamps and scene IDs, suitable for pasting into the existing Timestamped Scene Prompt Importer.

Use this exact structure:

Scene 001 | 00:00:00.000 - 00:00:04.200
Narration: [short aligned narration excerpt]
Asset IDs: CHAR_001, LOC_001, PROP_001
Image Prompt: [complete English image prompt]

Scene 002 | 00:00:04.200 - 00:00:08.750
Narration: [short aligned narration excerpt]
Asset IDs: CHAR_001, LOC_002
Image Prompt: [complete English image prompt]

Continue sequentially until the entire narration has been covered.

The copy-ready block must:

- Include all scenes in chronological order.
- Use the actual scene start/end timestamps.
- Use the existing importer-compatible `Scene ### | START - END` header.
- Include exactly one `Image Prompt:` field per scene.
- Avoid Markdown tables, commentary, or extra explanations inside the copy block.
- Keep explanatory notes and warnings outside the copy-ready block.
- Preserve scene IDs, timestamps, and prompt text when copying.
- Support copying all prompts with one click and copying individual or selected scenes.

- Never truncate the prompt block silently; if the UI cannot handle the complete output, provide clearly numbered copyable batches while preserving original scene IDs and timestamps.

G. REVIEW AND TIMESTAMP VALIDATION

Before finalizing the prompt block, display a review summary:

- Audio duration.
- Number of scenes created.
- Timestamp source: word-level alignment, forced alignment, or estimated.
- Scene coverage start/end.
- Gaps and overlaps.
- Unaligned or low-confidence narration sections.
- Script/audio mismatch warnings.
- Missing character/location/prop references.
- Scenes requiring manual review.

Provide editable start and end timestamps and an editable prompt for each scene. If a user edits a timestamp, re-run overlap/gap checks. Do not silently shift later scenes unless the user confirms a ripple adjustment.

Allow the user to approve the prompt list and then use **COPY ALL PROMPTS** to copy the exact validated text block. The copied text must match what is visible in the copy-ready panel.

H. FAILURE AND FALLBACK BEHAVIOR

If no voiceover is uploaded, the application may analyze the story and draft prompts, but must not claim audio-synchronized timestamps. It may offer approximate scene boundaries only when clearly marked as estimates.

If the audio cannot be decoded or aligned, explain which capability is missing and allow:

- Manual timestamp entry.
- A user-provided timestamped transcript.
- Estimated timing based on audio duration, explicitly marked as estimated.

If the story or audio is too long for one processing request, process it in manageable chunks while preserving a single global timeline, continuous scene numbering, and consistent asset IDs. Merge the results in chronological order and validate the complete output before enabling **COPY ALL PROMPTS**.

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22. FINAL BUILD INSTRUCTIONS

Build the application in the current environment using its actual supported capabilities.

Prioritize working functionality over decorative complexity.

The interface should be visually polished, but the core workflow must remain simple:

PASTE STORY + UPLOAD VOICEOVER → ANALYZE STORY + AUDIO → GENERATE TIMESTAMPED IMAGE PROMPTS → COPY ALL PROMPTS / IMPORT → UPLOAD REFERENCES → GENERATE ALL IMAGES → REVIEW TIMELINE → APPLY MOTION → RENDER FULL VIDEO → DOWNLOAD.

Start by implementing the application shell and the complete five-scene test workflow.

Then connect each control to its real operation, validate its behavior, and report which features have been tested successfully and which require an external integration.

Do not stop after creating the interface.

Do not claim successful MP4 export until a playable output file has actually been produced.

The final objective is a practical, scalable Manhwa Video Production Studio that minimizes manual editing while maintaining strict narration synchronization, character consistency, timestamp accuracy, and professional visual quality.